Exact Quantum Values and Permutation-Blind Maximizers in Cyclic Bell Inequalities

Sharp operator bounds, equality structure, and limits of Bell-value randomness certification

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Status and attribution. This is an unrefereed, AI-assisted manuscript. Perito, D'Avino, Jung, Mironowicz, Acín, and Augusiak introduced the two cyclic Bell families, supplied canonical strategies and lower bounds, conjectured the first reduced value, and SOS-proved the second. The present paper proves the first upper bound, strengthens it to commuting operators, and constructs alternative exact maximizers. It does not claim external review or challenge the source's fixed-canonical-behavior numerical calculation.

Exact operator value. With $\omega = e^{2\pi i/d}$,

$$\mathcal{I}_d = \sum_{y=0}^{d-1} \mathrm{Re}[(A_0 + \omega^y A_1)B_y] .$$

For arbitrary unitaries satisfying only cross-party commutation,

$$\mathcal{I}_d \leq 2\csc\left(\frac{\pi}{2d}\right) I.$$

After imposing the order-d measurement relations, the source's finite strategy attains this value, hence

$$\beta_q = \beta_{qa} = \beta_{qc} = 2\csc\left(\frac{\pi}{2d}\right).$$

The Hermitian augmentation has value one more.

The load-bearing operator identity is, for the canonical polar decomposition $C = V|C|$ and a commuting unitary B,

$$\frac{|C| + |C^\dagger|}{2} - \mathrm{Re}(CB) = \frac{1}{2}P^\dagger P, \qquad P = |C^\dagger|^{1/2} - V|C|^{1/2}B.$$

It is valid for a partial isometry with a kernel; no inverse or unitary extension is used. Functional calculus reduces the remaining gap to

$$\sum_{y=0}^{d-1} |1 + \omega^y z| \leq 2\csc\frac{\pi}{2d}, \qquad \text{equality iff } z^d = (-1)^{d-1}.$$

Equality-phase permutations. Let z_k be the d scalar equality roots. For any permutation κ, put

$$A_0 = X, \qquad A_1 = X\,\mathrm{diag}(z_{\kappa_j}),$$

and pair the same order with every Bob polar-phase row. Exact product-one identities make all weighted shifts order-d observables with simple full spectrum. Every complex first-harmonic correlator in the Bell score is a symmetric sum of the unordered root data and is independent of κ. This is a sufficient maximizing family, not a classification of all maximizers.

Target distribution and guessing gap. Define the cyclic cumulative sequence

$$q_0 = 1, \qquad q_{j+1} = z_{\kappa_j} q_j, \qquad \widehat{q}_m = \sum_j q_j \omega^{mj}.$$

The designated joint-output distribution is

$$p_\kappa(a, b \mid 1, d) = \frac{|\widehat{q}_{-(a+b)}|^2}{d^3}.$$

Parseval makes both marginals uniform. The canonical order is Fourier-flat. For every $d \geq 4$, swapping the last two roots gives lag-two autocorrelation

$$|R_2| = 4 \sin(\pi/d) \sin(3\pi/d) > 0,$$

so the joint table is nonuniform. With one-dimensional Eve,

$$G(AB \mid 1, d, E) \geq \frac{1}{d^2} + \frac{2\sin(\pi/d)\sin(3\pi/d)}{d^2(d-1)} > \frac{1}{d^2}.$$

At $d = 4$, independent exact arithmetic over $\mathbb{Q}(\zeta_{16})$ gives probabilities $1/32$ and $3/32$, hence $G = 3/32 > 1/16$.

Second family. For $\widehat{B}_\ell = \sum_y \omega^{\ell y} B_y$, the source certificate is

$$dI - \mathcal{F}_d = \frac{1}{2d} \sum_\ell (d\lambda_\ell I - A_\ell \widehat{B}_\ell)^\dagger (d\lambda_\ell I - A_\ell \widehat{B}_\ell).$$

The permuted Bob cycles satisfy the exact compression $\widehat{B}_\ell = d\lambda_\ell D_\ell$. Choosing $A_\ell = \overline{D_\ell}$ annihilates every factor and gives the same target table. Under the source appendix's alternative dagger convention this table is merely relabeled by Bob outcome inversion.

Exact scope. The scalar maximum does not identify the canonical behavior or certify $2\log_2 d$ private global bits across the entire maximizing face. This does not show that the canonical strategy lacks maximal randomness. An SDP that fixes the complete canonical distribution imposes higher Fourier data that the permuted behavior does not satisfy. Passing from fixed full behavior to one scalar value would require uniqueness, sufficient self-testing, or extra observed correlators.

Focused checks. Review should concentrate on: (1) partial-isometry support and commuting functional calculus; (2) paired permutation/product hypotheses; (3) the target DFT and guessing estimate; (4) second-family Fourier phase, Alice conjugation, and SOS normalization; and (5) the scalar-value/full-behavior logic. The package includes a fresh claims ledger, dependency map, priority audit, hostile tests, and two independent exact $d = 4$ verifiers.

Replay. `cd cyclic_bell_exact_values_and_randomness && ./reproduce.sh`

Canonical page. Open the canonical paper page and complete artifact index.